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DATA DESCRIPTOR

Oil Palm Fruits Dataset in Plantations for Harvest Estimation Using Digital Census and Smartphone

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This article presents a dataset of oil palm Fresh Fruit Bunches (FFBs) images from commercial plantations in Central Kalimantan, Indonesia, focusing on five maturity stages: Unripe, Underripe, Ripe, Flower, and Abnormal. The data collection involved smartphone video recordings of unharvested trees from multiple angles under varying conditions. Video frames were extracted and expertly annotated using Computer Vision Annotation Tool (CVAT), with annotations exported in Common Objects in Context (COCO) format suitable for object detection tasks. It has 10,207 images in its training set, 2,896 in the validation set, and 1,400 in the test set, which are supplemented using data augmentation to handle class imbalance and increase variation. These images have real-world complications arising from partial visibility, low contrast, occlusion, and blurriness. It forms the basis that will support the development of deep learning models for detection and classification of FFB, particularly for monitoring of harvest times, yield prediction, and optimization of resources in plantation operations.

Background & Summary

Manual approaches to counting palm oil fruits are inherently slow, costly, and lack scalability, underscoring the need for automated system¹. The complexity arises from the diverse shapes and ripeness stages of palm oil FFBs, making precise visual detection challenging. Advancements in artificial intelligence, particularly machine vision-based technologies, offer a more efficient alternative to manual counting^{2,3}. However, existing datasets for FFB detection on plantation sites are limited due to the lack of comprehensive training data^{4,5}. Creating a plantation dataset is crucial for monitoring harvest times, predicting dates, optimizing resource management, and minimizing waste in harvest operations⁶. The necessity arises from the complexities of palm oil cultivation, with FFBs located at various heights, resulting in images of varying detail and quality due to distance constraints and occlusions. Many studies focus on harvested fruits; however, datasets captured in plantation environments often contain a limited number of images and are less effective in capturing the challenging conditions where fruits are frequently occluded, clustered, or overlapping. This novel dataset supports our research on developing deep learning models for detection and counting in plantations⁷, serving as the main dataset for training and validation to verify the possibility of automatically counting and identifying FFBs in plantation environments.

This dataset comprises images of oil palm FFBs captured directly taken from the trees in plantations located in Central Kalimantan, Indonesia. As illustrated in Fig. 1, the dataset focuses on categorizing oil palm fruits into five maturity stages: Unripe (fruits in the initial stage), Underripe (fruits in the transitional stage), Ripe (fruits ready for harvesting), Flower (representing the flowering stage), and Abnormal (fruit with atypical characteristics due to potential disease or irregularity). Oppositely, many datasets focus either on harvested fruits, few numbers of classes, or single FFB per photo, as noted in Table 1, this dataset captures FFBs in their natural unharvested state, within the plantation environment. This approach, while exposing challenges of occlusions, conditions all going well, deliberately mimics conditions that practitioners face in field conditions rather

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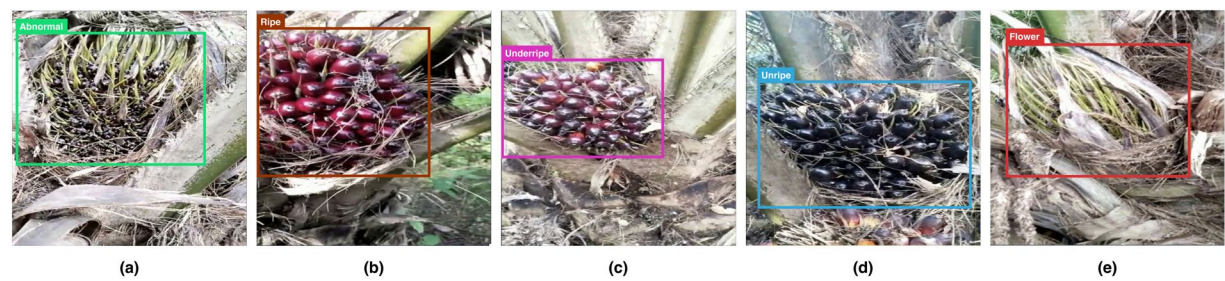


Fig. 1 Examples of annotated datasets: (a) Abnormal, (b) Ripe, (c) Under-ripe, (d) Unripe, (e) Flowering.

Dataset Type	Dataset Settings	FFB State	Number of Classes	Name of Classes	Total Images
Detection ¹¹	Single FFB per photo	Unharvested	4	1045 (Very Raw), 1045 (Underripe), 1045 (Ripe II), 1045 (Overripe)	4180
Detection ¹²	Oil palm plantation (close-up and distant views of the FFBs)	Unharvested	2	240 (Ripe) and 250 (Unripe)	490
Classification ¹³	Single FFB per photo	Harvested	3	39 (Unripe), 40 (Ripe), and 41 (Overripe)	120
Classification ¹⁴	Perpendicularly 1 meter above the ground	Unharvested	4	30 (Raw Bunch), 30 (Under Ripe Bunch), 60 (Ripe Bunch), 60 (Over Ripe Bunch)	180
Detection ¹⁵	Multiple harvested oil palm (but mostly one fruit per picture)	Harvested	6	1130 (Unripe), 1289 (Underripe), 1880 (Ripe), 1162 (Overripe), 473 (Empty Bunch), 1237 (Abnormal FFB)	7171
Classification ¹⁶	Single FFB per photo	Harvested	5	130 (Unripe), 131 (Underripe), 130 (Ripe), 130 (Overripe), 130 (Abnormal)	653
Detection (Our Dataset)	Oil palm plantation (close-up and distant views of the FFBs), unharvested	Unharvested	5	2989 (Ripe), 1803 (Underripe), 7453 (Unripe), 1554 (Flower), 704 (Abnormal)	14503

Table 1. Prior FFB studies and proposed dataset.

than ideal, controlled conditions. To ensure reliability under these challenging conditions, the dataset has been pre-processed and validated by an expert in oil palm fruits to enable the creation of applications that detect the maturity levels of FFBs and estimate the number of fruit bunches on oil palm trees using different methods and approaches.

Methods

The creation of this dataset involved several stages, as illustrated in Fig. 2. Initially, video data of oil palm FFBs was collected from commercial oil palm plantations in Central Kalimantan province, Indonesia, in 2023. Using a smartphone video camera, footage was captured under various conditions, including different tree heights and recording times. The data collection process entailed recording videos by rotating around each tree to capture all fruit images from multiple angles.

The captured videos were then extracted frame by frame into images using the CVAT (Computer Vision Annotation Tool) application. This facilitated the identification and annotation of fruit objects, which were categorized into five classes of ripeness levels: unripe, underripe, ripe, flower, and abnormal. Bounding boxes were drawn around each fruit and oil palm flower in the images. An expert, seasoned in quality control, defined the fruit ripeness levels to ensure accurate labeling and annotation. The annotation results were exported in COCO format, making the dataset suitable for processing using the YOLO (You Only Look Once) method.

Next, the dataset was split into three subsets with a ratio of 70% training data, 20% validation data, and 10% testing data. To address class imbalance and enhance data variation, data augmentation techniques were applied using the Albumentations library⁸. The augmentation methods included random cropping (80% probability), horizontal and vertical flips (50% probability), 90° rotations, brightness and contrast adjustments with a limit of 20 (50% probability), addition of Gaussian noise, RGB shift, and HSV shift (50% probability), both with a shift limit value of 20. Data balancing was achieved through over-sampling and under-sampling methods to equalize the number of objects across classes.

Data Records

The dataset for this study was gathered from commercial oil palm plantations in Central Kalimantan Province, Indonesia, by recording videos at 30 frames per second inside the plantation⁹. The data used in this study were acquired with a smartphone to take video recordings of FFB on oil palm trees. The trees selected were 1–3 meters in height, and the recording process involved circling each tree to capture all visible FFB. Sample frames from these videos can be seen in Fig. 3. In total, 440 videos were collected in this study, where the durations are ranging from 8 seconds to 1 minute and 31 seconds. The recording resolution is 320 × 640 pixels (which is in portrait orientation). All those videos are stored in the MP4 format.

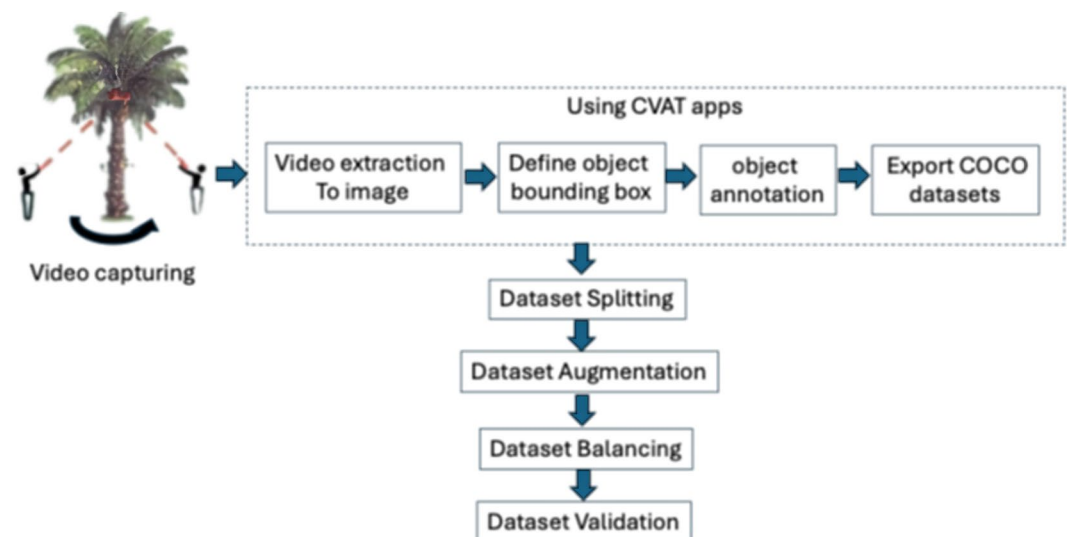


Fig. 2 The stages of dataset development.

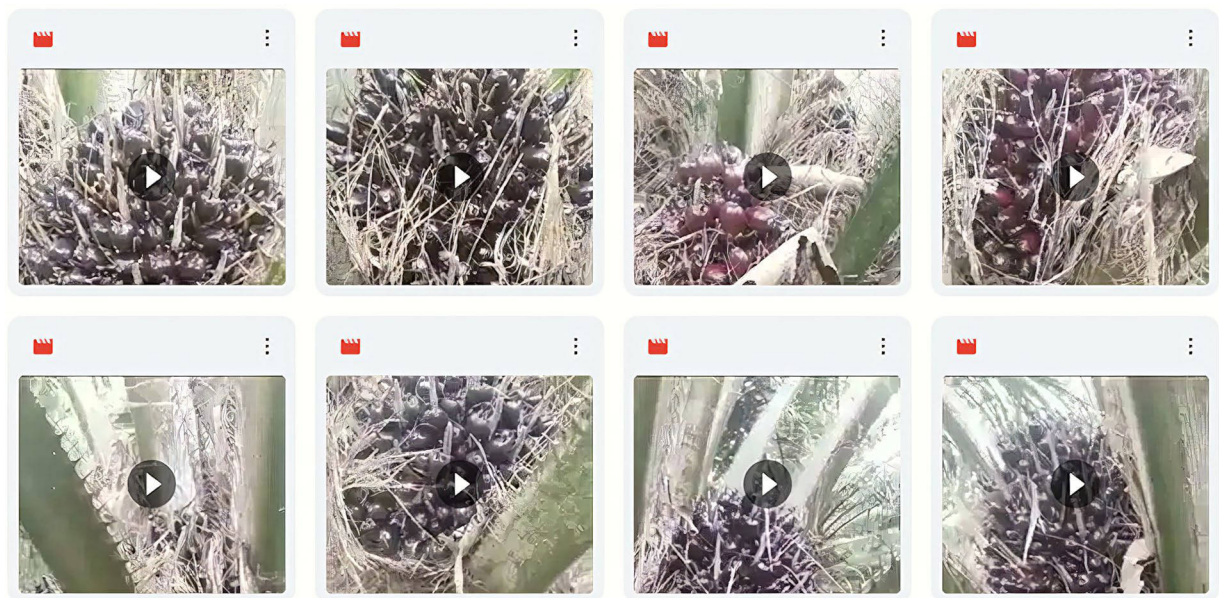


Fig. 3 Example of video data collection results for oil palm FFB.

Initially, the dataset was split into training, validation, and test sets with a 70:20:10 ratio. However, through augmentation techniques, the dataset's size increased significantly, resulting in the distribution shown in Table 2. The “Unripe” class has the most substantial representation, with 4,990 images in the training set, 1,699 in the validation set, and 764 in the test set. This is primarily due to the nature of the dataset, which focuses on FFBs in the plantation; naturally, there are more unripe fruits since most ripe ones are harvested promptly, leaving a larger number of unripe fruits to be documented. Conversely, the “Abnormal” class has the fewest images, reflecting its lower occurrence and the specific nature of these instances. Overall, the dataset consists of 10,207 training images, 2,896 validation images, and 1,400 test images, providing a well-rounded and extensive set of data to train, validate, and test object detection models¹⁰.

The dataset presents numerous challenges for the detection and counting of FFBs due to complexities inherent in real-world conditions. These challenges in Fig. 4 include partial visibility, where objects are only partially visible within the image frame (refer to a); low contrast, resulting from lighting conditions that cause classes to resemble one another (refer to b); occlusion, where objects are blocked by other objects (refer to c); and blurriness, which reduces perceptual clarity (refer to d). These complexities mirror the actual conditions found in oil palm plantations, emphasizing the need for sophisticated models capable of accurately recognizing and classifying FFBs in their natural state on the trees.

Class	TRAIN	Validation	Test
Ripe	2354	443	192
Underripe	1324	298	181
Unripe	4990	1699	764
Flower	1121	251	182
Abnormal	418	205	81
Total	10207	2896	1400

Table 2. Distribution of pre-processing dataset results.

	Before augmentation		After augmentation	
	mAP_0.5	mAP_0.5:0.95	mAP_0.5	mAP_0.5:0.95
YOLOv8s	71.9	47.1	74.5	48.0

Table 3. Effect of image augmentation on model performance.

	YOLOv8n	YOLOv8s
Max Epoch	100	100
Batch size	64	16
Height	640	640
Width	640	640
Channels	3	3
Learning Rate	0.01	0.01
Optimizer	SGD	SGD
Momentum	0.9	0.9
Workers	8	8
Seed	10	10

Table 4. Hyper-parameters of YOLOv8 models for dataset validation.

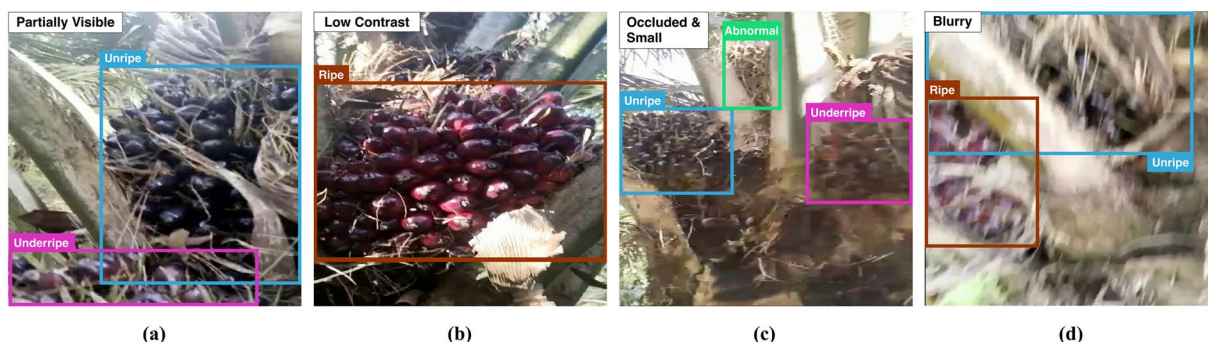


Fig. 4 Dataset complexity.

This dataset is suitable for developing applications to detect FFB maturity levels in plantations, calculating the number of fruits on oil palm trees, and conducting research in machine learning and deep learning for object detection and classification. It is particularly useful for application development for mobile devices, robots, and embedded systems.

Technical Validation

Before deploying the models, the effectiveness of image augmentation used in this dataset needs to be evaluated. While augmentation techniques were applied, there still remained significant class imbalances, especially between the “Unripe” and “Abnormal” classes. This is because FFBs in these developmental stages are inherently uncommon in plantations, hence setting a limit to how much augmentation can be done without risking overfitting of the models on repetitive, slightly modified versions of a few images. Ablation study in comparing the model trained with and without augmentations to assess its impacts despite the class disparities was presented in Table 3. The results confirmed the benefits of augmentation, evidenced by improvements in both mAP_0.5 and mAP_0.5:0.95 scores.

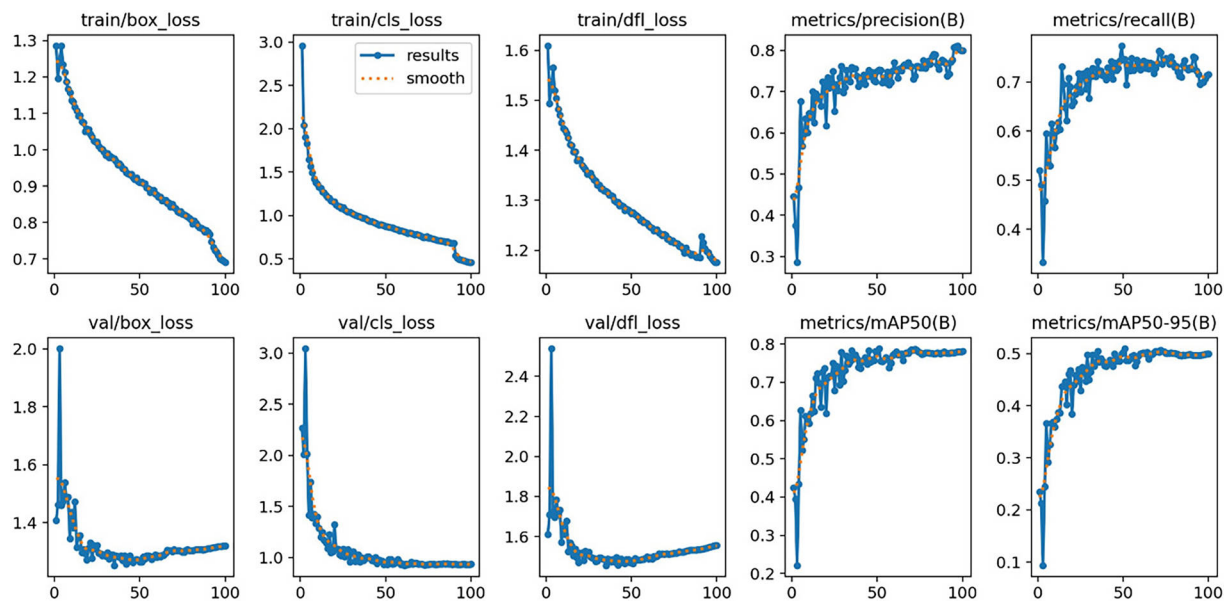


Fig. 5 YOLOV8n model training performance.

	YOLOV8n	YOLOV8s
GPU used	9.6gb	4.5GB
mAP_0.5	0.735	0.745
mAP_0.5:0.95	0.480	0.480
Precision	0.725	0.738
Recall	0.711	0.719
F1 score	0.718	0.728
FPS	90	36
Train box loss	0.690	0.684
Train class loss	0.460	0.563
Train object loss	1.175	1.135
Val. box loss	1.251	1.260
Val. class loss	0.924	0.923
Val. object loss	1.456	1.490
Size model	6MB	21 mb

Table 5. Training, Validation, and Testing Results of YOLOv8 models.

Ultimately, the dataset was validated by the development of a palm oil FFB detection and counting model using YOLOv8, which was then implemented on a smartphone. Technical validation was performed by training YOLOv8n and YOLOv8s models with identical core hyperparameters for fair comparison, as shown in Table 4. Both models were configured to run for a maximum of 100 epochs, with batch sizes of 64 for YOLOv8n and 16 for YOLOv8s. The input dimensions were set to 640×640 pixels with 3 input channels (RGB images). The learning rate was 0.01, using Stochastic Gradient Descent (SGD) optimizer with a momentum of 0.9. Eight workers were used for data loading, and a random seed of 10 ensured reproducibility.

The training performance of the YOLOv8n model demonstrated smooth convergence across all monitored metrics (see Fig. 5). The training box loss, class loss, and distribution focal loss (DFL) consistently declined, indicating effective learning as the number of epochs increased. Validation losses followed a similar trend, starting slightly higher due to differences in data complexity between training and validation sets. As can be seen from Table 5, YOLOv8n achieved a precision and recall of 0.725 and 0.711, respectively, at the final epoch, while its mAP scores show an overall improvement with minor variabilities.

On the other hand, the YOLOv8s model, in Fig. 6, had a similar learning trend but was automatically stopped at epoch 87 since there was no visible improvement for 50 epochs. While the training losses had a smooth, downwards trend, the validation losses were quite erratic, with box loss and DFL loss being especially erratic in terms of fluctuations in the validation data. In spite of that, all metrics for the YOLOv8s model were marginally better, where precision and recall achieved 0.738 and 0.719, respectively, while mAP_0.5 was 0.745, edging out the 0.735 for YOLOv8n. However, the consistent mAP_0.5:0.95 results obtained in both models show that the high precision of YOLOv8s did not help improve performance across the entire scale of IoU thresholds.

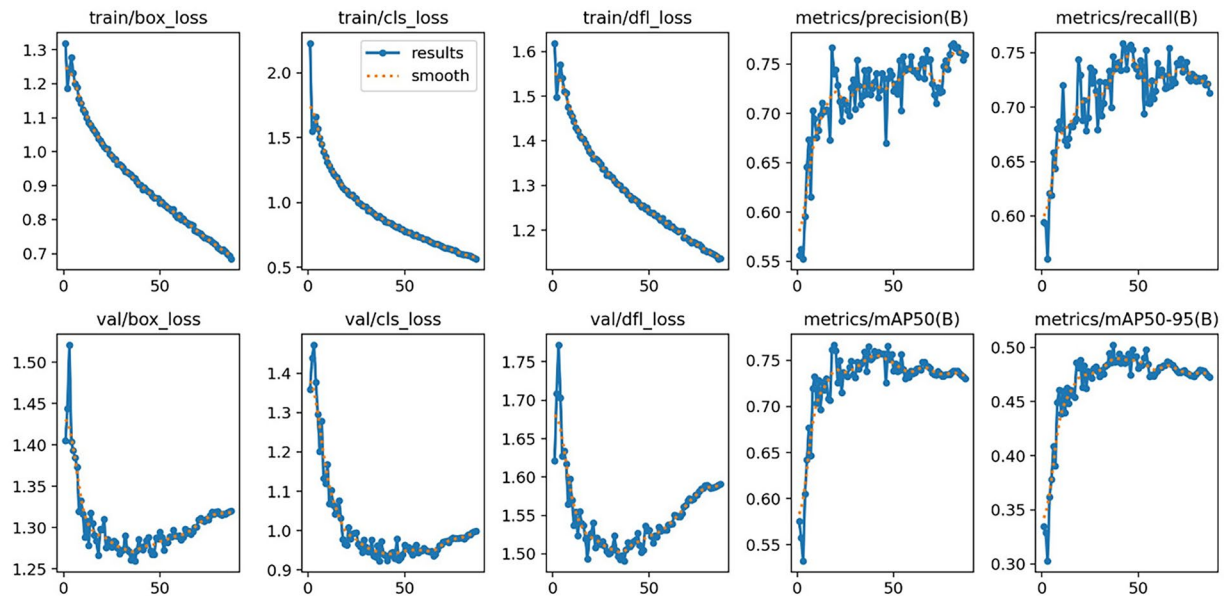


Fig. 6 YOLOV8s model training performance.

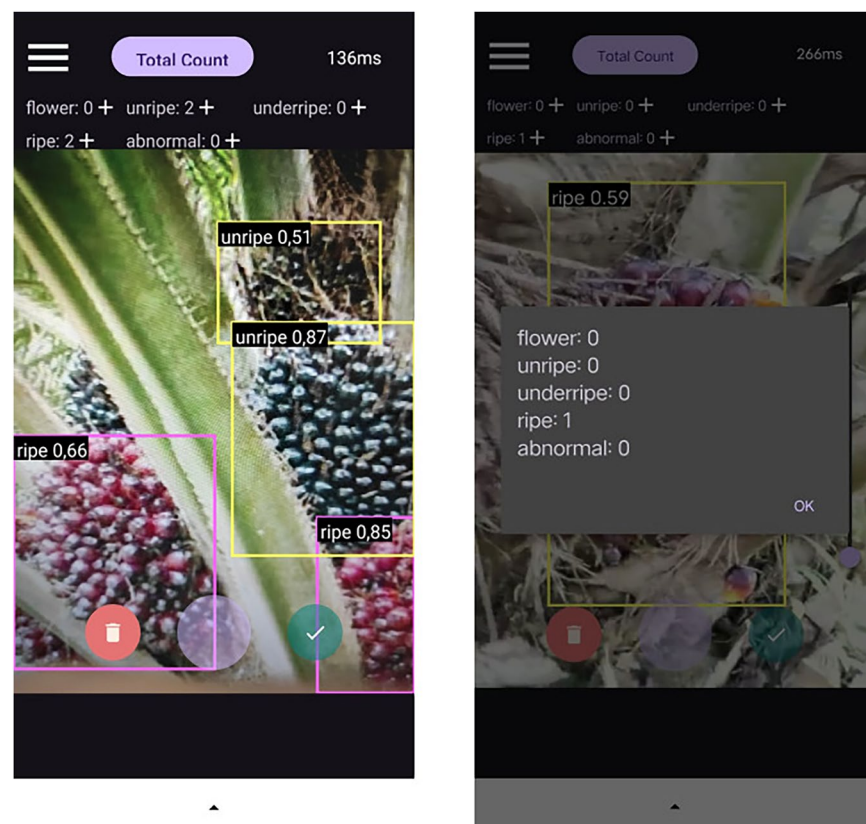


Fig. 7 Mobile-based e-census prototype of oil palm FFB.

The other important aspect of technical validation involves model efficiency, referring to the use of the GPU and the size of the model. YOLOv8n has a GPU usage of 9.6GB, thus being much lightweight in contrast to YOLOv8s, which used 4.5GB of GPU memory. On the other hand, YOLOv8s had a much lower value for frames per second (FPS). Specifically, YOLOv8s ran in 36 FPS compared to YOLOv8n's 90 FPS. On the opposite side, YOLOv8n has a smaller size at 6MB, whereas the YOLOv8s size is at 21MB. These metrics suggested a tradeoff between the size of the model and the processing speed. In particular, the higher GPU memory usage of YOLOv8n during training, despite its smaller model size, can be largely explained by the larger batch size

used. The batch size is one of the key drivers of GPU memory usage, as it directly controls the number of images being processed in parallel. That means that YOLOv8s uses less GPU memory to train with a smaller batch size.

Lastly, Fig. 7 shows the application of the YOLOv8 model in a mobile-based e-census prototype specifically designed for FFBs detection and classification. This real-world implementation serves as a necessary validation mechanism to assert the model's performance outside laboratory settings and into practical applications. This prototype can detect and classify the various maturity stages of FFBs directly at the site. The interface in Fig. 7 gives real-time feedback on the recognized objects, showing the classification confidence for each FFB with its respective class label (e.g., ripe, underripe, unripe, flower, or abnormal). The ability of the system to process images very fast, with measured response times in milliseconds, emphasizes the efficiency of the model in practical scenarios. In the left part of Fig. 7, there are quite a number of FFBs detected and classified, together with confidence scores for each detected instance. Whereas, in the right part of Fig. 7, an example with fewer detected FFBs is shown, and the results are summarized in a pop-up interface. This proves that the system can adapt to changing conditions, such as, the number of visible FFBs or differences in image quality.

Usage Notes

This dataset has the following limitations:

1. This dataset was collected on oil palm trees with a height of between 1–3 meters so that this dataset does not represent every image of oil palm FFB in plantations that have tree heights above it.
2. This dataset was collected using a smartphone that has resolution limitations so that the right application for this dataset is an application that uses data input from a smartphone or similar embedded applications.

Code availability

The software used in this study includes CVAT (<https://www.cvat.ai>) for converting video data into frames and performing annotation, including labeling and creating bounding boxes for each object. Albumentations (<https://albumentations.ai/>) was employed for image augmentation to enhance the training dataset. Additionally, the Ultralytics repository (<https://github.com/ultralytics/ultralytics>) was utilized for implementing YOLOv8 and customizing its configuration to suit the specific requirements of the research.

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Author contributions

Suharjito: Responsible for research design, investigation, securing funding, data collection and preparation, and drafting the manuscript; Martinus Grady Naftali: Handled data curation, wrote the initial draft, developed software, and validated the dataset; Gregory Hugo: Worked on software development, data labeling, and dataset validation; Muhammad Reza: Focused on data labeling and dataset validation; Muhammad Asrol: handled data validation and verification; Ditdit Nugeraha Utama: worked on software evaluation and testing. All author agreed with final manuscript.

Competing interest

The authors state that they have no conflicts of interest or personal relationships that could have impacted the findings presented in this paper.

Additional information

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